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Association of Glycaemic Control With Lipid Profiles Among Adults With Type 2 Diabetes Mellitus in Indonesia: A Retrospective Cross-Sectional Study

Abstract

Background. Altered lipid metabolism is frequently found in type 2 diabetes mellitus and may intensify cardiovascular risk. This study assessed whether routinely measured lipid fractions differed between adults with adequate and inadequate glycaemic control.

Methods. A retrospective cross-sectional review was conducted using consecutive anonymised medical records of 120 adults with type 2 diabetes mellitus treated at an Indonesian tertiary hospital from January 2022 to December 2023. Glycaemic status was defined by glycated haemoglobin, with values below 7% classified as controlled and values of 7% or above classified as uncontrolled. Total cholesterol, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol and triglycerides were analysed using independent-samples t tests or the Mann-Whitney U test according to data distribution.

Results. Among the 120 included records, 77 patients (64.2%) were in the uncontrolled-glycaemia group. Mean total cholesterol was higher in this group than in the controlled group (210.86 ± 51.64 vs. 179.23 ± 43.24 mg/dL; mean difference, 31.63 mg/dL; 95% confidence interval, 13.23 to 50.03; $p = 0.001$). Triglyceride values were also greater in the uncontrolled group (median 156, range 57-828 mg/dL) than in the controlled group (median 128, range 49-385 mg/dL; $p = 0.013$). Low-density lipoprotein cholesterol showed a higher numerical value without statistical significance ($p = 0.077$), whereas high-density lipoprotein cholesterol was comparable between groups ($p = 0.585$).

Conclusion. In this hospital-based sample, uncontrolled glycaemia was linked to higher total cholesterol and triglyceride levels. Evaluating HbA1c together with lipid measurements may assist clinicians in recognising patients who need closer cardiometabolic-risk management.

Keywords: Diabetes Mellitus, Type 2; Hemoglobin A; Cholesterol; Cholesterol, LDL; Cholesterol, HDL; Triglycerides; Dyslipidemias; Cross-Sectional Studies.

Introduction

Type 2 diabetes mellitus (T2DM) continues to impose a substantial public-health burden in Indonesia. The International Diabetes Federation reported that 19.5 million Indonesian adults were affected in 2021, and national health-survey findings indicated an upward trend in diagnosed diabetes between 2013 and 2018.^{1,2}

Disordered lipid metabolism is a common accompaniment of T2DM. Its usual phenotype is not limited to high low-density lipoprotein cholesterol (LDL-C); it also involves increased triglycerides, accumulation of triglyceride-rich remnant particles, lower high-density lipoprotein cholesterol (HDL-C), and qualitative changes in LDL particles that enhance atherogenic potential.^{3,4,12-16}

Glycated haemoglobin (HbA1c) is used in routine practice as a marker of average glycaemic exposure over recent months. For many non-pregnant adults, a target below 7% is commonly applied, although the appropriate target may vary according to age, comorbidity, hypoglycaemia risk, and other clinical factors. Management of lipids remains a parallel priority because cardiovascular disease is a major cause of illness and death in patients with T2DM.^{5,6}

The reported relationship between HbA1c and specific lipid parameters has not been uniform. Several observational studies have found stronger and more consistent differences for total cholesterol and triglycerides than for LDL-C or HDL-C, but findings vary by population, treatment exposure, and health-care setting.^{7-9,17-20}

Indonesian hospital-based data on this issue remain limited, especially in samples that exclude patients receiving lipid-lowering medication. Therefore, this study compared total cholesterol, LDL-C, HDL-C, and triglyceride levels between adults with controlled and uncontrolled T2DM in an Indonesian hospital setting.

Methods

Study design and setting. This study was a retrospective cross-sectional analysis of anonymised secondary medical-record data from a tertiary hospital in Indonesia. The eligible source records were dated from January 2022 through December 2023. Reporting was guided by the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement.¹⁰

Participants and data source. Records were included consecutively when they belonged to adults with documented T2DM and contained both an HbA1c measurement and a complete lipid profile. In the source records, T2DM was documented on the basis of HbA1c $\geq 6.5\%$ and/or fasting plasma glucose ≥ 126 mg/dL. Records were excluded when patients had received lipid-lowering therapy, including statins, fibrates, niacin, or ezetimibe, or when chronic kidney disease, hyperthyroidism, or liver cirrhosis was documented. After applying these criteria, 120 records were analysed.

Variables. The main exposure was glycaemic status, grouped as controlled when HbA1c was $< 7\%$ and uncontrolled when HbA1c was $\geq 7\%$. The outcome variables were total cholesterol, LDL-C, HDL-C, and triglycerides, all recorded in mg/dL. Age, sex, and body mass index (BMI) were used to describe the sample. BMI categories followed Asian cut-off values: underweight (< 18.5 kg/m²), normal weight (18.5-22.9 kg/m²), overweight (23.0-24.9 kg/m²), obesity class I (25.0-29.9 kg/m²), and obesity class II (≥ 30.0 kg/m²).^{5,11}

Statistical analysis. SPSS version 24 was used for analysis. Data distribution was assessed before selecting the comparative test. Normally distributed lipid values were summarised as mean $\pm$ standard deviation (SD) and compared using independent-samples t tests. Triglyceride values were non-normally distributed; therefore, they were summarised as median (range) and compared using the Mann-Whitney U test. Categorical data were reported as number and percentage. A two-sided p-value < 0.05 was considered statistically significant. Because the available dataset allowed only unadjusted group comparisons, the results are presented as associations and not as evidence of causality.

Ethical considerations. The study received approval from the relevant institutional ethics committee, and permission for the use of hospital data was granted by the participating institution. Only de-identified secondary medical-record data were used. There was no direct patient contact, and confidentiality was maintained throughout the study.

Results

The analysis included 120 adults with T2DM. The largest age group was 45-59 years ($n = 62$, 51.7%). Most participants were female ($n = 73$, 60.8%), and obesity class I was the most

frequent BMI category (n = 49, 40.8%). Overall, 77 participants (64.2%) were classified as having uncontrolled glycaemia (Table 1).

Table 1: Demographic and clinical characteristics of adults with type 2 diabetes mellitus in Indonesia (N = 120).

Characteristic	n (%)
Age, years	
19-44	10 (8.3)
45-59	62 (51.7)
≥ 60	48 (40.0)
Sex	
Male	47 (39.2)
Female	73 (60.8)
Body mass index, kg/m²	
Underweight (< 18.5)	7 (5.8)
Normal weight (18.5-22.9)	14 (11.7)
Overweight (23.0-24.9)	24 (20.0)
Obesity class I (25.0-29.9)	49 (40.8)
Obesity class II (≥ 30.0)	26 (21.7)
Glycaemic control	
Controlled (HbA1c < 7%)	43 (35.8)
Uncontrolled (HbA1c ≥ 7%)	77 (64.2)

HbA1c: glycated haemoglobin.

When lipid values were compared by glycaemic-control group, total cholesterol and triglycerides were higher among patients with uncontrolled glycaemia (Table 2). LDL-C was also higher in absolute terms, but the confidence interval crossed zero and the p-value did not meet the threshold for statistical significance. HDL-C was similar in both groups.

Table 2: Serum lipid profiles by glycaemic-control status among adults with type 2 diabetes mellitus in Indonesia.

Lipid profile	Controlled (n = 43)	Uncontrolled (n = 77)	²² Difference (95% CI)	p-value
LDL-C, mg/dL	109.70 ± 38.66	122.62 ± 37.72	12.92 (-1.43 to 27.27)	0.077 ^a
HDL-C, mg/dL	45.86 ± 13.66	47.29 ± 13.68	1.43 (-3.72 to 6.58)	0.585 ^a
Total cholesterol, mg/dL	179.23 ± 43.24	210.86 ± 51.64	31.63 (13.23 to 50.03)	0.001 ^a
Triglycerides, mg/dL	128 (49-385)	156 (57-828)	Not applicable	0.013 ^b

Values are mean ± SD, except triglycerides, which are median (range). Difference was calculated as uncontrolled minus controlled for normally distributed variables. LDL-C: low-density lipoprotein cholesterol; HDL-C: high-density lipoprotein cholesterol; CI: confidence interval. ^a Independent-samples t test. ^b Mann-Whitney U test.

Discussion

This retrospective cross-sectional analysis showed that patients with uncontrolled glycaemia had higher total cholesterol and triglyceride levels than those whose HbA1c was below 7%. LDL-C was higher numerically but was not statistically different, and HDL-C values were almost comparable between groups. These results suggest that poor glycaemic status and lipid abnormalities may cluster in routine clinical practice, although the design of the study does not allow temporal or causal conclusions.

The findings are in line with several studies from resource-limited and middle-income settings, where less favourable glycaemic control has often been accompanied by higher total cholesterol or triglyceride concentrations. Nevertheless, the literature does not show a uniform pattern for all

lipid fractions. Variation in treatment exposure, adiposity, population characteristics, and laboratory timing may contribute to these differences.^{7-9,17,18}

The association observed in this study is biologically plausible. Insulin resistance and sustained hyperglycaemia can stimulate lipolysis in adipose tissue and increase hepatic very-low-density lipoprotein output. These processes favour hypertriglyceridaemia and an increased concentration of remnant particles. In addition, glycation and oxidative modification can change lipoprotein metabolism and may increase atherogenicity.^{3,4,12-16}

The non-significant results for LDL-C and HDL-C should not be read as absence of cardiovascular risk. In T2DM, conventional LDL-C concentration may underestimate the contribution of small dense LDL, remnants, and other triglyceride-rich particles. HDL-C concentration is also affected by factors that were not captured in this review, including sex, smoking, diet, physical activity, adiposity, and medication history.^{6,13,19,20}

Several limitations should be considered. The retrospective and cross-sectional design prevents assessment of the temporal sequence between glycaemia and lipid changes. The study was conducted in a single hospital, so the findings may not represent all patients with T2DM in Indonesia. The analysis did not adjust for diabetes duration, antihyperglycaemic treatment, diet, physical activity, smoking, socioeconomic status, or other potential confounders. The timing of HbA1c and lipid-profile testing depended on available clinical records and may not have been identical for all patients. Exclusion of patients using lipid-lowering drugs reduced direct drug-related effects on lipid values, but it may also limit generalisation to the broader T2DM population. Even so, the study adds local hospital-based evidence using clinical measurements that are routinely available.

Conclusion

In adults with T2DM treated in an Indonesian hospital setting, uncontrolled glycaemia was associated with higher total cholesterol and triglyceride levels. LDL-C and HDL-C did not differ significantly between the glycaemic-control groups. Routine interpretation of HbA1c together with lipid parameters may provide a more complete view of cardiometabolic risk. Larger prospective studies with adjustment for clinical and lifestyle factors are needed to confirm these findings.

5 Conflict of interest

The authors declare no conflict of interest.

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